

FINAL PROJECT OUTCOME / OBJECTIVE DOCUMENT

Smart Belongings Recovery System for Campus

Major Project – I (01CE0716)

Submitted By

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1. Project Overview

The **Smart Belongings Recovery System for Campus** is a **web application** developed to simplify and secure reporting, tracking, identification, and recovery of lost belongings within a college campus.

Unlike a conventional lost-and-found system, it provides a **Smart Gadget Registration and QR Tag mechanism**. Students can register smart gadgets in advance and associate them with a unique QR Tag. If a registered gadget is found after being lost, the QR Tag can be scanned to initiate recovery without directly revealing the owner's personal information. OTP-based ownership verification provides an additional security layer before recovery.

2. Project Objectives

- Develop a centralized web application for reporting and recovering lost/found belongings.
- Allow students to report items with details, location, date, and images.
- Provide status tracking throughout recovery.
- Allow students to **register smart gadgets** in advance.
- Provide **QR Tags for registered smart gadgets** for quick identification.
- Protect owner's personal information during initial QR identification.
- Implement **OTP-based ownership verification** before recovery.
- Provide in-app and email notifications.
- Provide an Admin Panel for users, items, devices, reports, and recovery.
- Improve security using email verification, password hashing, sessions, and role-based access control.

3. Final Project Outcome

The final outcome is a functional **web application** consisting of Student and Admin modules.

Student Module: Registration/Login, Email Verification, Dashboard, Lost/Found Reporting, Image Upload, My Items, Status Tracking, Smart Gadget Registration, QR Tag, QR Scan, OTP Recovery Verification, Notifications, and Profile Management.

Admin Module: Admin Dashboard, User Management, Lost/Found Item Management, Item Status Updates, Smart Device Management, Reports/Complaints, Recovery Operations, and Notifications.

4. Special Smart Gadget Recovery Feature

The main distinguishing feature is the **Smart Gadget Recovery System**. A student can register a smart gadget using brand, model, serial number, and device image. The system associates the registered gadget with a unique QR

Token/QR Tag.

Recovery flow:

Register Smart Gadget → Assign QR Tag → Gadget Gets Lost → Finder Scans QR Tag → Recovery Process Starts → OTP Ownership Verification → Secure Recovery.

5. Technology Used

Frontend: HTML5, CSS3, JavaScript, Bootstrap

Backend: PHP

Database: MySQL

Database Management: phpMyAdmin

Server Environment: XAMPP (Apache + MySQL)

Application Type: Web Application

6. Security Features

- Email verification
- Bcrypt password hashing
- Session-based authentication
- Role-based access control
- OTP-based ownership verification
- Privacy-preserving QR identification
- Unique QR tokens for registered devices

7. Software Prototype Evidence

The developed web application includes Landing Page, Registration, Login, Email Verification, Student Dashboard, Report Lost/Found Item, My Items, My Device, QR Scan, Notifications, My Profile, Admin Dashboard, Manage Items, Update Item Status, Manage Users, and Reports/Complaints screens.

8. Project Workflow

1. Student registers an account.
2. Email verification is completed.
3. Student logs in.
4. Student reports a lost/found item or registers a smart gadget.
5. Information is stored in the database.
6. Admin reviews and manages it.
7. Item/device status is updated during recovery.
8. Notifications are sent to the concerned student.
9. QR scanning initiates smart-gadget recovery.
10. OTP verification confirms ownership before secure recovery.

9. Expected Impact / Benefits

- Centralized campus lost-and-found platform
- Faster reporting and recovery
- Better tracking of belongings
- Easier identification of registered smart gadgets
- Secure ownership verification
- Reduced exposure of personal information
- Better communication through notifications
- Centralized administration
- Low-cost campus-specific solution

10. Future Scope

Android/iOS application, SMS notifications, advanced administrator analytics, campus ID/LMS integration, AI-based image matching, cloud deployment, and multi-campus support.

11. Project Status

Project Status: Academic Major Project – I

Application Type: Web Application

Development Environment: XAMPP Local Server

Target Environment: College/Campus

12. Prototype Access / Verification

GitHub Repository: <https://github.com/azimkadivar30/Smart-Belongings-Recovery-System-for-Campus>

Live Demo: Not publicly deployed; current prototype runs in the XAMPP local-server environment.

The GitHub repository provides access to the project source code and project documentation. The repository is public for academic evaluation and faculty access.